EAST TEXAS A&M UNIVERSITY

CSCI 352.61E
DIGITAL FORENSICS

CARVING/RECOVERING A USB IMAGE

using scalpel

PREPARE A USB IMAGE FOR FILE CARVING

Download the zipped USB image

```
┌──(root💀kali80)-[~/carvingLab]
└─# wget  -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_image/120M.7z

┌──(root💀kali80)-[~/carvingLab]
└─# ls -l 120M.7z
-rw-r--r-- 1 root root 36720470 Sep 24 09:03 120M.7z
```

Compute hashes

```
┌──(root💀kali80)-[~/carvingLab]
└─# hashdeep -c md5,sha1 120M.7z
%%%% HASHDEEP-1.0
%%%% size,md5,sha1,filename
## Invoked from: /root/carvingLab
## # hashdeep -c md5,sha1 120M.7z
##
36720470,dfe7b5424e54cd1bf50d5df47aceeb3c,2810745018afaa2da31dc17a8eb590fca66eeef7,/root/carvingLab/120M.7z
```

size md5 sha1

List the content of the zipped file

```
┌──(root💀kali80)-[~/carvingLab]
└─# 7z l 120M.7z

7-Zip [64] 16.02 : Copyright (c) 1999-2016 Igor Pavlov : 2016-05-21
p7zip Version 16.02 (locale=en_US.UTF-8,Utf16=on,HugeFiles=on,64 bits,2 CPUs Intel(R) Xe
on(R) Gold 6226R CPU @ 2.90GHz (50650),ASM,AES-NI)

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Listing archive: 120M.7z

--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

   Date      Time    Attr         Size   Compressed  Name
------------------- ----- ------------ ------------  ------------------------
2021-09-22 22:59:31 D...A            0            0  120M
2021-09-22 22:59:31 ....A    124780544     36720256  120M/usb_fat_carving.001
2021-09-22 22:59:32 ....A         1685               120M/usb_fat_carving.001.txt
------------------- ----- ------------ ------------  ------------------------
2021-09-22 22:59:32          124782229     36720256  2 files, 1 folders
```

content of the zip

List the content of the zipped file

```
┌──(root💀kali80)-[~/carvingLab]
└─# 7z e 120M.7z

7-Zip [64] 16.02 : Copyright (c) 1999-2016 Igor Pavlov : 2016-05-21
p7zip Version 16.02 (locale=en_US.UTF-8,Utf16=on,HugeFiles=on,64 bits,2 CPUs Intel(R) Xe
on(R) Gold 6226R CPU @ 2.90GHz (50650),ASM,AES-NI)

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Extracting archive: 120M.7z
--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

Everything is Ok

Folders: 1
Files: 2
Size:       124782229
Compressed: 36720470
```

e : Extract files from archive (without using directory names)

Verify the hashes

```
┌──(root💀kali80)-[~/carvingLab]
└─# hashdeep -c md5,sha1 usb_fat_carving.001
%%%% HASHDEEP-1.0
%%%% size,md5,sha1,filename
## Invoked from: /root/carvingLab
## # hashdeep -c md5,sha1 usb_fat_carving.001
##
124780544,ba4a1d0ba49f4a6667b00a3b3e85e604,bcc2d49fd49c9521ecb1739f6542c6bf327375ef,/root/carvingLab/usb_fat_carving.001

┌──(root💀kali80)-[~/carvingLab]
└─# cat usb_fat_carving.001.txt | grep "checksum"
 MD5 checksum:     ba4a1d0ba49f4a6667b00a3b3e85e604
 SHA1 checksum:    bcc2d49fd49c9521ecb1739f6542c6bf327375ef
 MD5 checksum:     ba4a1d0ba49f4a6667b00a3b3e85e604 : verified
 SHA1 checksum:    bcc2d49fd49c9521ecb1739f6542c6bf327375ef : verified
```

EXAM THE CONTENT OF THE USB

Display partitions

```
┌──(root💀kali80)-[~/carvingLab]
└─# fdisk -l usb_fat_carving.001
Disk usb_fat_carving.001: 119 MiB, 124780544 bytes, 243712 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa1159a00

Device                 Boot Start    End Sectors   Size Id Type
usb_fat_carving.001p1  *      128 243711  243584 118.9M  e W95 FAT16 (LBA)
```

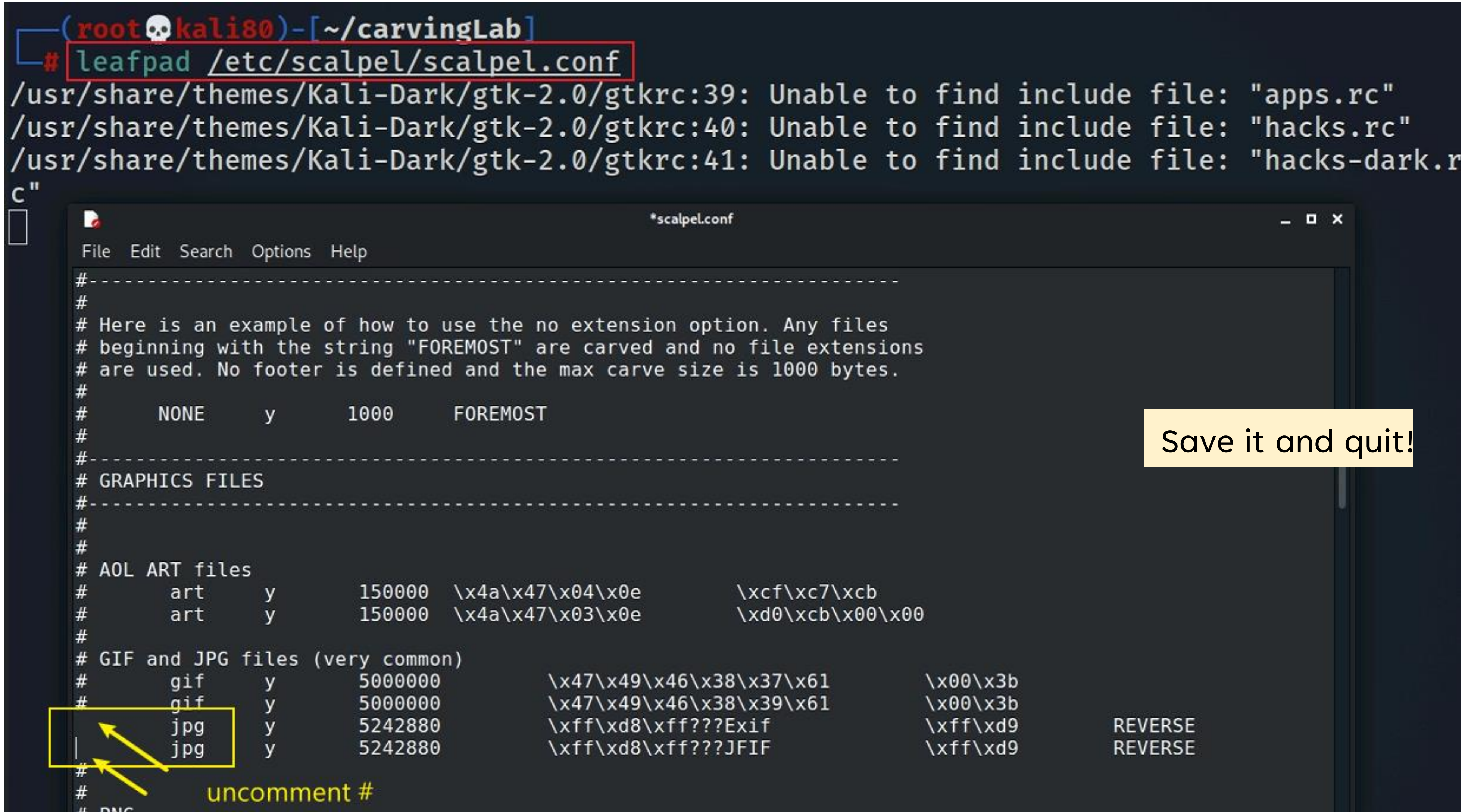
Find deleted files

```
┌──(root💀kali80)-[~/carvingLab]
└─# fls -o 128 usb_fat_carving.001
r/r 3:  USB         (Volume Label Entry)
d/d 6:  System Volume Information
r/r * 7:        _est
r/r 10: .dropbox.device
d/d * 13:       old_File_Carving_files
r/r * 15:       B_ub_poe4.bmp
r/r * 18:       B_zoom-eubie-mono.bmp
r/r * 21:       Ballardlab8.java
r/r * 24:       brittLab10.java
r/r * 27:       DO_example.doc
r/r * 30:       DO_example2.doc
r/r * 33:       G_BuiltForThis.gif
r/r * 35:       G_zoom-sc.gif
r/r * 37:       H_Form.html
r/r * 39:       H_hello.html
r/r * 41:       J_ub_law.jpg
r/r * 44:       J_ub_night.jpg
r/r * 47:       nps-2008-jean_outlook.pst
r/r * 50:       P_CAS-zoom-6.png
r/r * 53:       P_MSB_1_zoom.png
r/r * 57:       pd_Evidence_search_techniques.pdf
r/r * 61:       pd_Forensic_Report_Template.pdf
r/r * 64:       pp_Number_Systems.pptx
r/r * 67:       pp_one_page.pptx
r/r 69: readme.docx
r/r 70: readme.txt
r/r * 71:       _tf_1.rtf
r/r * 74:       T_Eubie-iphone-5-8.tiff
r/r * 78:       T_youknowus-iphone-5-8.tiff
r/r * 81:       W_axloadcomplete.wav
r/r * 84:       W_speaker_test_sound.wav
r/r * 86:       Z_file.zip
r/r * 88:       江莎莎行踪轨迹.doc
r/r * 91:       nps-2008-jean_outlook.pst
v/v 3889667:    $MBR
v/v 3889668:    $FAT1
v/v 3889669:    $FAT2
V/V 3889670:    $OrphanFiles
```

deleted

deleted

DECIDE WHICH FILE TYPES NEED TO CARVE



Show help

```
┌──(root💀kali80)-[~/carvingLab]
└─# scalpel -h
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.
Carves files from a disk image based on file headers and footers.

Usage: scalpel [-b] [-c <config file>] [-d] [-h|V] [-i <file>]
                 [-m blocksize] [-n] [-o <outputdir>] [-O num] [-q clustersize]
                 [-r] [-s num] [-t <blockmap file>] [-u] [-v]
                 <imgfile> [<imgfile>] ...

-b   Carve files even if defined footers aren't discovered within
     maximum carve size for file type [foremost 0.69 compat mode].
-c   Choose configuration file.
-d   Generate header/footer database; will bypass certain optimizations
     and discover all footers, so performance suffers.  Doesn't affect
     the set of files carved.  **EXPERIMENTAL**
-h   Print this help message and exit.
-i   Read names of disk images from specified file.
-m   Generate/update carve coverage blockmap file.  The first 32bit
     unsigned int in the file identifies the block size. Thereafter
     each 32bit unsigned int entry in the blockmap file corresponds
     to one block in the image file.  Each entry counts how many
     carved files contain this block. Requires more memory and
     disk.  **EXPERIMENTAL**
-n   Don't add extensions to extracted files.
-o   Set output directory for carved files.
-O   Don't organize carved files by type. Default is to organize carved files
     into subdirectories.
-p   Perform image file preview; audit log indicates which files
     would have been carved, but no files are actually carved.
-q   Carve only when header is cluster-aligned.
-r   Find only first of overlapping headers/footers [foremost 0.69 compat mode].
-s   Skip n bytes in each disk image before carving.
-t   Set directory for coverage blockmap.  **EXPERIMENTAL**
-u   Use carve coverage blockmap when carving.  Carve only sections
     of the image whose entries in the blockmap are 0.  These areas
     are treated as contiguous regions.  **EXPERIMENTAL**
```

CARVING THE USB IMAGE

```
┌──(root💀kali80)-[~/carvingLab]
└─# scalpel usb_fat_carving.001 -o output                                        1 ×
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/root/carvingLab/usb_fat_carving.001"

Image file pass 1/2.
usb_fat_carving.001: 100.0% |***********************************|  119.0 MB    00:00 ETA
Allocating work queues...
Work queues allocation complete. Building carve lists...
Carve lists built.  Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x45\x78\x69\x66" and footer "\xff\xd9" --> 8 f
iles
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" --> 9 f
iles
Carving files from image.
Image file pass 2/2.
usb_fat_carving.001: 100.0% |***********************************|  119.0 MB    00:00 ETA
Processing of image file complete. Cleaning up...
Done.
Scalpel is done, files carved = 17, elapsed = 0 seconds.
```

Show carved files

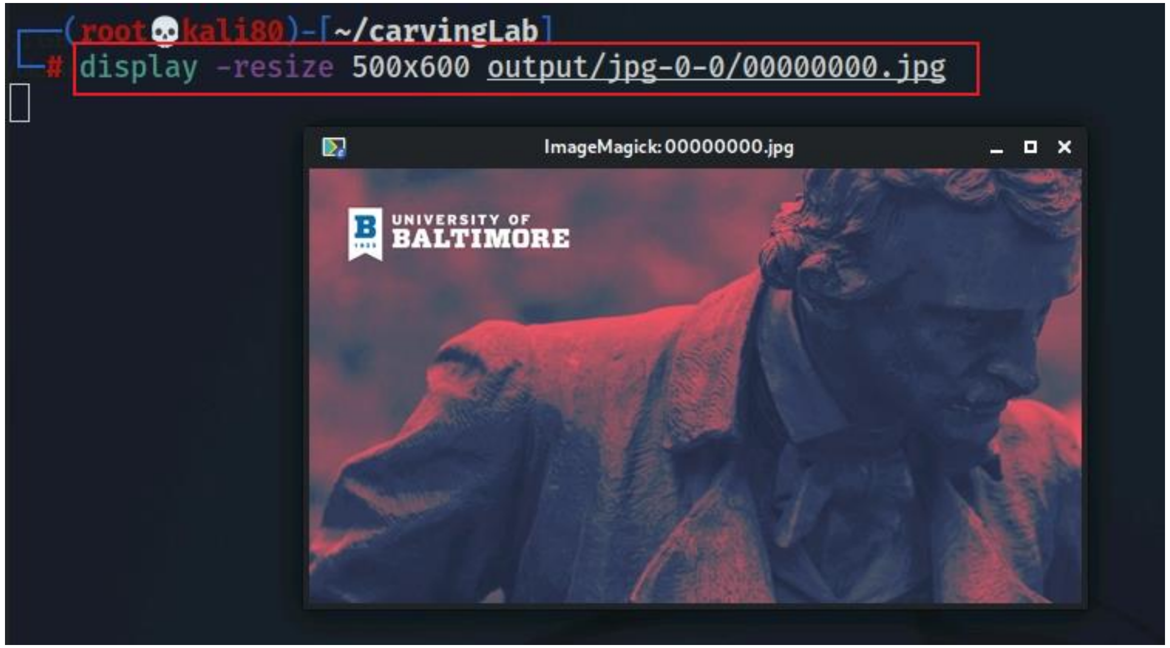
```
┌──(root💀kali80)-[~/carvingLab]
└─# tree output
output
├── audit.txt
├── jpg-0-0
│   ├── 00000000.jpg
│   ├── 00000001.jpg
│   ├── 00000002.jpg
│   ├── 00000003.jpg
│   ├── 00000004.jpg
│   ├── 00000005.jpg
│   ├── 00000006.jpg
│   └── 00000007.jpg
└── jpg-1-0
    ├── 00000008.jpg
    ├── 00000009.jpg
    ├── 00000010.jpg
    ├── 00000011.jpg
    ├── 00000012.jpg
    ├── 00000013.jpg
    ├── 00000014.jpg
    ├── 00000015.jpg
    └── 00000016.jpg

2 directories, 18 files
```

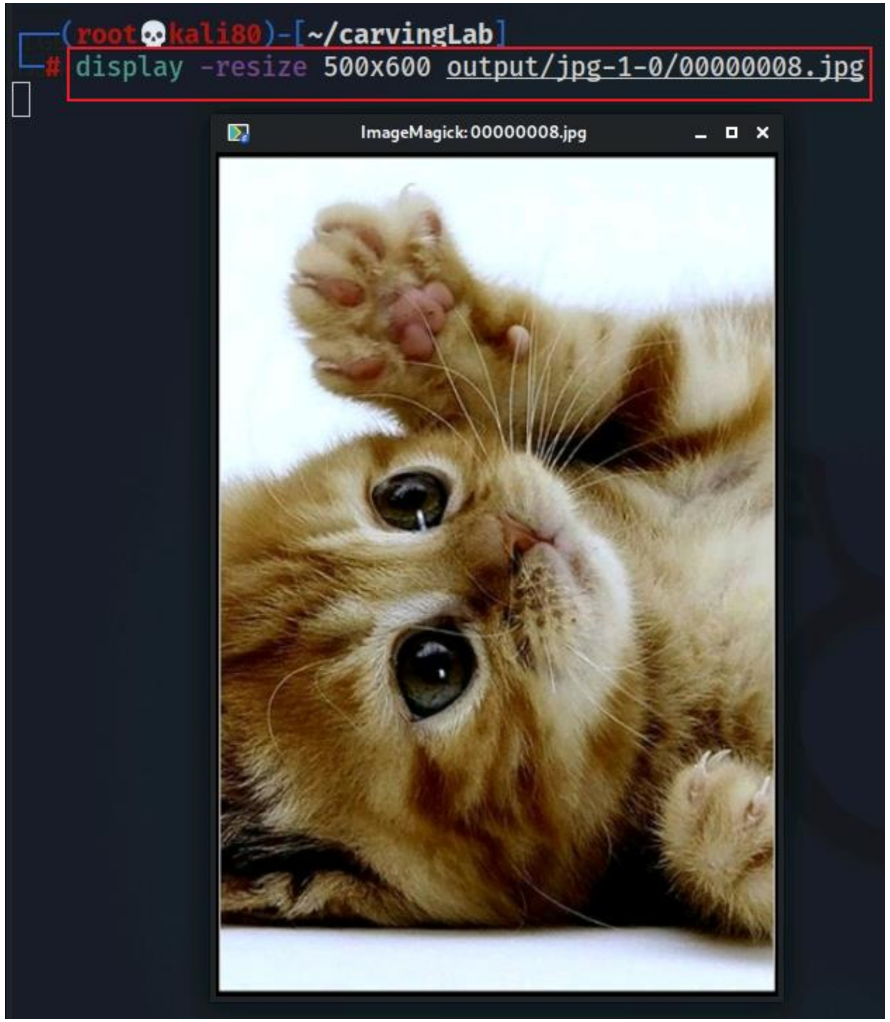
carved files

Show audit log

```
┌──(root💀kali80)-[~/carvingLab]
└─# cat output/audit.txt

Scalpel version 1.60 audit file
Started at Fri Sep 24 11:20:43 2021
Command line:
scalpel usb_fat_carving.001 -o output

Output directory: /root/carvingLab/output
Configuration file: /etc/scalpel/scalpel.conf

Opening target "H="

The following files were carved:
File                  Start                     Chop          Length        Extracted From
00000010.jpg        3796992                     NO           3148500        usb_fat_carving.001
00000009.jpg         425822                     NO           4875802        usb_fat_carving.001
00000008.jpg         335872                     NO           4965752        usb_fat_carving.001
00000002.jpg        6938624                     NO              6868        usb_fat_carving.001
00000001.jpg        5285888                     NO           1659604        usb_fat_carving.001
00000000.jpg        3897344                     NO           3048148        usb_fat_carving.001
00000004.jpg       15427584                     NO           4223822        usb_fat_carving.001
00000003.jpg       12881920                     NO           4256310        usb_fat_carving.001
00000006.jpg       19638272                     NO           5229050        usb_fat_carving.001
00000005.jpg       17121280                     NO           4248530        usb_fat_carving.001
00000007.jpg       21358592                     NO           5183267        usb_fat_carving.001
00000016.jpg       36671488                     NO           2889262        usb_fat_carving.001
00000015.jpg       36571136                     NO           2989614        usb_fat_carving.001
00000014.jpg       36393610                     NO           3167140        usb_fat_carving.001
00000013.jpg       36352448                     NO           3208302        usb_fat_carving.001
00000012.jpg       35590996                     NO           3969754        usb_fat_carving.001
00000011.jpg       35350242                     NO           4210508        usb_fat_carving.001


Completed at Fri Sep 24 11:20:43 2021
```

DISPLAY TWO CARVED JPG IMAGE





SUMMARY

- File header and tail signatures are used to identify file types
- The content of a file is enclosed in header and tail signatures
- Many tools can used for carving file content
- Note the file names/metadata can't be retrieved

THANK YOU

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